Given a function $f$, we know how to write its Taylor series centered at a point $a$, and we know how to find its interval of convergence. However, we still do not know whether the series actually converges to $f$. Last time we proved the Maclaurin series for the function

$$g(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0 & x = 0, \end{cases}$$

is identically zero and therefore does not converge to $g$ on its interval of convergence. Thus, there exist functions which are not equal to the sum of their Taylor series. Under what conditions, then, will a function be equal to the sum of its Taylor series? Clearly, a necessary condition is that all the derivatives of $f$ exist at $a$. Convergence of the Taylor series to $f$ on its interval of convergence is then the statement that

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k := \lim_{n\to\infty} \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(x) \tag{1}$$

for all $x$ in the interval of convergence of the series. Each partial sum $T_n(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$ is a polynomial in $x$ of degree $n$ called the *nth-degree Taylor polynomial of $f$ at $a$*.

For any function $f$ and non-negative integer $n$, we can write $f(x) = T_n(x) + R_n(x)$, where $R_n(x) := f(x) - T_n(x)$ is called the *remainder*. The absolute value of the remainder is the error made in approximating $f(x)$ by $T_n(x)$.

1.) Prove that $f(x) = \lim_{n\to\infty} T_n(x)$ if and only if $\lim_{n\to\infty} R_n(x) = 0$.

To calculate $\lim_{n\to\infty} R_n(x)$, we need an expression for $R_n$ in terms of $f$. This is provided by *Taylor's Theorem*:

**Theorem. (Taylor's Theorem)** If $f$ has $(n+1)$ continuous derivatives on an open interval $I$ containing $a$, then for all $x$ in $I$,

$$f(x) = T_n(x) + R_n(x),$$

where $T_n$ is the $n$th-order Taylor polynomial for $f$ with center $a$ and the remainder is

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}, \tag{2}$$

for some point $c$ between $x$ and $a$. Equation (2) is called the *Lagrange form* of the remainder.

We will now prove Taylor's Theorem.

(a) Show that the Fundamental Theorem of Calculus can be written in the form

$$f(x) = f(a) + \int_a^x f'(t_1)\, dt_1. \tag{3}$$

(b) By (a), $f'(t_1) = f'(a) + \int_a^{t_1} f''(t_2)\, dt_2$. Substitute this into (3) to show that

$$f(x) = f(a) + f'(a)(x-a) + \int_a^x \int_a^{t_1} f''(t_2)\, dt_2\, dt_1. \tag{4}$$

(c) By (a) again, $f''(t_2) = f''(a) + \int_a^{t_2} f^{(3)}(t_3)\, dt_3$. Substitute this into (5) to show that

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \int_a^x \int_a^{t_1} \int_a^{t_2} f^{(3)}(t_3)\, dt_3\, dt_2\, dt_1. \tag{5}$$

Continuing in this way, we obtain

$$f(x) = \underbrace{\sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k}_{T_n(x)} + \underbrace{\int_a^x \int_a^{t_1} \cdots \int_a^{t_n} f^{(n+1)}(t_{n+1})\, dt_{n+1} \cdots dt_2\, dt_1}_{R_n(x)}. \quad (6)$$

**(Optional: Prove** (6) **by induction on** $n$ **on a separate sheet of paper.)**

(d) We will now establish the Lagrange form of the remainder. Consider first the case where $x > a$. Claim: $f^{(n+1)}$ has a minimum value $m$ and a maximum value $M$ on $[a,x]$. Explain why this is true.

(e) By definition of $m$ and $M$, we have

$$m \le f^{(n+1)}(t_{n+1}) \le M$$

for all $t_{n+1} \in [a,x]$. Integrate this double inequality $n+1$ times to show that

$$m\frac{(x-a)^{n+1}}{(n+1)!} \le R_n(x) \le M\frac{(x-a)^{n+1}}{(n+1)!}. \quad (7)$$

(*Hint:* Recall that $f(x) \ge g(x)$ for $a \le x \le b$ implies $\int_a^b f(x)\, dx \ge \int_a^b g(x)\, dx$.)

(f) Claim: The function $t \mapsto f^{(n+1)}(t)\frac{(x-a)^{n+1}}{(n+1)!}$ is continuous. Explain why this is true.

(g) Claim: The function in part (f) attains a maximum and a minimum value on $[a,x]$. Explain why this is true.

(h) Find the extreme values of the function in part (f) on $[a,x]$. (*Hint*: They depend on $m$ and $M$, the extreme values of $f^{(n+1)}$.)

(i) Claim: It follows that there exists $c \in (a,x)$ such that $R_n(x) = f^{(n+1)}(c)\frac{(x-a)^{n+1}}{(n+1)!}$, completing the proof for the case where $a < x$. Explain why the claim is true. If $x < a$ and $n$

is odd, the same proof works. If $x < a$ and $n$ is even, then $(x-a)^{n+1} < 0$ and the same proof works after reversing some inequalities. **(Optional: Work out the details.)** □

**Example:** The Maclaurin series for $f(x) = e^x$ converges to $f(x)$ for all real $x$.

**Proof.** The Maclaurin series for $e^x$ is $\sum_{k=0}^{\infty} \frac{x^k}{k!}$, which converges for all real $x$. By Taylor's Theorem, the remainder is

$$\begin{aligned} R_n(x) &= \frac{f^{(n+1)}(c)}{(n+1)!}x^{n+1} \\ &= \frac{e^c}{(n+1)!}x^{n+1} \quad (\text{since } \frac{d^n}{dx^n}e^x = e^x) \end{aligned}$$

for some $c$ in $[0,x]$. To show that $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$, we need to show that $\lim_{n\to\infty} R_n(x) = 0$. Note that $c$ depends on $n$, so we can't pull $e^c$ out of $\lim_{n\to\infty} R_n(x)$. Without knowing the exact form of $c(n)$, we cannot compute the limit. However, for all $n$ we have $0 \le c \le x$, and since the exponential is an increasing function, this implies that $1 = e^0 \le e^c \le e^x$, and therefore

$$|R_n(x)| = \frac{e^c}{(n+1)!}|x|^{n+1} \le \frac{e^x}{(n+1)!}|x|^{n+1},$$

or equivalently

$$-\frac{e^x}{(n+1)!}|x|^{n+1} \le |R_n(x)| \le \frac{e^x}{(n+1)!}|x|^{n+1}.$$

Note that

$$\lim_{n\to\infty} \frac{e^x}{(n+1)!}|x|^{n+1} = e^x \lim_{n\to\infty} \frac{1}{(n+1)!}|x|^{n+1} = 0,$$

since the Maclaurin series $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ converges for all real $x$ and hence $\lim_{k\to\infty} \frac{x^k}{k!} = 0$. It therefore follows from the Squeeze Theorem that $\lim_{n\to\infty} |R_n(x)| = 0$ and hence $\lim_{n\to\infty} R_n(x) = 0$ for all real $x$, which proves that $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$.

2.) Prove that $\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}$ and $\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$ for all real $x$.

3.) Use the Maclaurin series for $\cos(x)$ to compute $\cos(5^\circ)$ correct to five decimal places.

4.) Compute $\sin(50^\circ)$ correct to five decimal places in two ways:

(a) Using the Maclaurin series for $\sin(x)$.

(b) Using the Taylor series for $\sin(x)$ centered at $\frac{\pi}{4}$.

Which of these converges to $\sin(50^\circ)$ faster? Why?